31338 Network Servers
Lab 1b: System startup, runlevels and log files

Aims

1. To be able to boot a machine into single user mode, using the bootloader
2. To explore and modify the system startup procedures in Linux
3. To be able to change runlevels
4. To become familiar with the system log files

Task 1: Boot single-user shell only, using the boot loader

Boot up your virtual machine (if already running, reboot it). You will see the Virtual machine BIOS screen for a few seconds. Then the GRUB2 boot loader will begin running and show a message: "Press any key . . ." Interrupt the boot process by pressing a key when you see this message. It might take several attempts to get your timing right. You should now see the GRUB boot loader screen. Follow the process below to instruct GRUB to boot in single user mode.

1. The GRUB2 boot loader shows the kernels that are available to boot. You can install as many as you like by later editing the /etc/grub2.cfg file. Highlight the desired kernel then press the 'E' key to edit the boot commands.
2. The screen will change to one with several lines. Move the cursor with the arrow keys to the line that starts "linux....". We want to pass a new option to the kernel, to tell it to boot with just a simple Unix shell.
3. On this line, look for where it says "ro" as a word by itself. You need to:
   Delete the word "`ro`"
   In its place, instead type: `rw init=/sysroot/bin/sh`
4. When you are sure you have typed it correctly, press Ctrl-X to boot the machine with your new boot parameters.

The machine will now boot and provide you with a root (single-user) shell without asking for a password. Changing "ro" (read-only) into "rw" (read-write) means that you will be able to edit files if you need to as the filesystem will be mounted in a writable mode. Note that the actual files of your machine are found inside the `/sysroot` folder when you are in single-user mode.

Single-user mode is mainly used for system maintenance, when you need to be sure that no-one else is using the system, and that a minimum number of processes are running. Document the process of booting into single-user mode using GRUB2.

Also note that this didn't require you to enter any passwords, so it is also the way to do password recovery on a Linux system using GRUB2. The downside is that if an attacker has physical access to your machine, they can do this too. It is possible to secure the GRUB2 bootloader so you cannot do what we have just done without entering a password. There is also a 'rescue mode' that still gives you a single-user environment, but **does** require the root password.

Task 2: Explore and modify system startup scripts

From single-user mode, we want to now exit and go back to graphical mode. The simplest way is to use the "reboot" command.

There are different modes the system can boot in. Single-user mode is one. Multi-user mode without graphics is another. Graphical mode is a third. Previously these were called "runlevels". In the latest releases they are called "targets".

Use the command `systemctl get-default` to check the current default mode. It should show "graphical.target", indicating that the default boot mode of your system is graphical mode. You can change it with "systemctl set-default"

The main targets are:

- `emergency.target` – used for emergency system recovery (a bit like single user shell)
- `rescue.target` – rescue mode – also for system recovery – requires root password
- `multi-user.target` – multi-user mode with no graphical login
- `graphical.target` – full graphical mode

Try now changing to multi-user mode, with the command: `systemctl isolate multi-user.target`
Document what happens in your learning journal.

Try changing back with: `systemctl isolate graphical.target`

Finally, try emergency.target and rescue.target. Document what happens in your learning journal.

systemctl is the command that manages the startup process, including which services start at boot time. We have just seen "targets", now let's explore "services". Run:

```
systemctl list-unit-files
```

It shows a list of all "units" – units can be targets or services or a few other kinds (we just focus on targets and services). For services, it will show you whether they are enabled or disabled.

Is the "sshd" service enabled or disabled by default? What about the "httpd" service?

Another way to check whether a single service is enabled or disabled is to use the 'is-enabled' command to systemctl, e.g.

```
systemctl is-enabled sshd
```

Enabled/disabled means whether the service will/won't start automatically when the machine boots. We can also check whether a service is active (currently running) or inactive.

```
systemctl is-active sshd
```

We can make it active ("start") or inactive ("stop"). We can make it start at boot ("enable") or not start at boot ("disable").

Try:

```
systemctl start <servicename>
systemctl stop <servicename>
systemctl enable <servicename>
systemctl disable <servicename>
```

Verify what's happening using "is-enabled" and "is-active" and document in your journal.

Google to learn more about using systemctl to manage the system startup process.

Task 3: Examine system log information

Run the `dmesg` command to view the contents of the kernel ring buffer. What kinds of information do you see? What about if you run the command `journalctl --dmesg`?

Also examine the file `/var/log/messages`. What kind of messages do you find in this file?

What about `/var/log/secure`?

HINT: sometimes you won't be able to read files – you should check their permissions (try ls –l). You may have to `su` as root to change permissions (try chmod).

Now experiment with the "journalctl" command. Try the following commands, document them in your learning journal and if you're not sure what they're doing, look them up.

```
journalctl --since "10 minutes ago"
journalctl --since "2020-01-01" --until "yesterday"
journalctl -u multi-user.target
journalctl -u sshd.service
journalctl -u sshd.service --since "10 minutes ago"
journalctl -p err
journalctl -p warning
journalctl -p err --since "1 hour ago"
```

<u>REMEMBER to note anything new into your learning journal!</u>